

FOOD ORDERING BEHAVIOUR AND CONSUMER TRENDS: A STRUCTURED ANALYSIS OF CHOICES AND HABITS

Project Report

Submitted By

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Technology Used: Tableau Public

Project Type: Business Intelligence & Data Visualization

Project Links

Live Dashboard Demo: <https://devisrikoppakula-24.github.io/SmartBridge-project-dashboard/>

GitHub Repository: <https://github.com/devisrikoppakula-24/SmartBridge-project-dashboard.git>

Abstract

The rapid growth of online food delivery services has resulted in the generation of large amounts of customer and operational data. Businesses often struggle to extract meaningful insights from this data to improve customer satisfaction and operational efficiency.

This project presents an interactive Tableau dashboard that analyzes food ordering behavior using a structured dataset containing order details, customer ratings, delivery information, meal types, cuisines, restaurant categories, and repeat orders. The dashboard transforms raw data into visual insights that enable stakeholders to identify trends, evaluate performance, and support strategic business decisions.

The project focuses on analyzing customer preferences, revenue generation, city-wise demand, delivery performance, and restaurant popularity through interactive charts and KPI indicators.

1. Introduction

The online food delivery industry has experienced tremendous growth over the past decade. Companies receive thousands of orders daily, making it essential to analyze customer behavior and operational performance.

Business Intelligence (BI) tools such as Tableau help organizations visualize complex datasets and convert them into meaningful insights.

This project demonstrates how Tableau can be used to analyze food ordering data and develop interactive dashboards that support business decision-making.

2. Problem Statement

Food delivery companies collect a significant amount of operational and customer data. However, identifying customer preferences, delivery efficiency, and revenue patterns from raw data is difficult.

Without proper visualization and analysis, organizations may face challenges such as:

- Difficulty understanding customer behavior
- Inefficient delivery operations
- Poor marketing decisions
- Low customer retention
- Ineffective resource allocation

The project addresses these issues through interactive data visualization.

3. Objectives

The objectives of this project are:

- Analyze customer ordering behavior.
- Identify the most popular cuisines.
- Compare city-wise order volumes.
- Calculate business KPIs.
- Study customer meal preferences.
- Analyze delivery performance.
- Evaluate repeat customer behavior.
- Develop an interactive Tableau dashboard.

4. Scope of the Project

The project helps food delivery businesses to:

- Understand customer preferences.
- Monitor operational performance.
- Improve delivery efficiency.
- Increase customer satisfaction.

- Support marketing decisions.
- Enhance business growth.

5. Tools and Technologies Used

Tool	Purpose
Tableau Public	Dashboard Development
Microsoft Excel	Data Storage
GitHub	Project Repository
CSV Dataset	Data Source

6. Dataset Description

The dataset contains approximately 50,000 food orders collected from multiple cities.

Attributes

Column	Description
Order ID	Unique Order Identifier
City	Customer Location
Cuisine	Food Category
Meal Type	Breakfast, Lunch, Dinner, Snacks
Restaurant Type	Restaurant Category
Order Value	Total Bill Amount
Delivery Fee	Delivery Charges
Rating Given	Customer Rating
Time Taken To Order	Delivery Time
Is Repeat Order	Customer Repeat Status

7. Data Preprocessing

Before creating visualizations, the dataset was cleaned and prepared. The preprocessing steps included:

- Loading the dataset into Tableau.
- Verifying data types.
- Removing duplicate records.
- Checking missing values.
- Formatting numeric fields.
- Creating calculated fields where necessary.
- Aggregating measures for KPI calculations.

8. Dashboard Development

The dashboard was developed using Tableau Public. The development process included the following steps:

- Step 1 — Import Dataset
- Step 2 — Data Cleaning
- Step 3 — Create KPIs
- Step 4 — Create Charts
- Step 5 — Apply Filters
- Step 6 — Design Dashboard
- Step 7 — Publish Dashboard

9. Dashboard Components

KPIs

- Total Orders
- Total Revenue
- Average Order Value
- Average Delivery Fee
- Average Rating

Visualizations

Orders by City

Displays city-wise order count. Purpose: identify cities with maximum demand.

Revenue by Cuisine

Shows revenue generated by each cuisine. Purpose: identify high-performing cuisines.

Meal Type Distribution

Displays customer meal preferences. Purpose: understand demand during different meal periods.

Average Rating by Cuisine

Compares customer ratings. Purpose: identify highly rated cuisines.

Delivery Time by City

Analyzes average delivery time. Purpose: evaluate operational efficiency.

Restaurant Type Distribution

Shows order volume across restaurant categories. Purpose: identify popular restaurant types.

Repeat Orders

Displays percentage of repeat customers. Purpose: measure customer loyalty.

10. Interactive Filters

The dashboard includes interactive filters:

- City
- Cuisine
- Meal Type
- Restaurant Type

These filters allow users to explore the data dynamically.

11. Key Insights

Some important observations from the dashboard include:

- Certain cities generate the highest number of food orders.
- Popular cuisines contribute significantly to revenue.
- Lunch and Dinner account for the majority of customer orders.
- Customer ratings vary across cuisines.
- Delivery performance differs by city.
- Some restaurant categories receive more customer orders.
- Repeat customers represent a significant portion of total orders.

12. Business Benefits

The dashboard helps businesses:

- Improve operational planning.
- Optimize delivery performance.
- Increase customer satisfaction.
- Identify profitable cuisines.
- Enhance marketing strategies.
- Support business decision-making.

13. Challenges Faced

During the project, several challenges were encountered:

- Data preprocessing.
- KPI creation.
- Dashboard layout design.
- Chart formatting.
- Interactive filter implementation.
- Publishing to Tableau Public.

14. Future Enhancements

Future improvements may include:

- Real-time data integration.
- Predictive analytics using Machine Learning.
- Monthly and yearly sales trend analysis.
- Customer segmentation.
- Mobile dashboard optimization.
- Advanced forecasting techniques.

15. Conclusion

The Food Ordering Behaviour and Consumer Trends dashboard successfully transforms raw food ordering data into meaningful business insights using Tableau.

The interactive dashboard enables users to analyze customer behavior, delivery performance, cuisine popularity, restaurant preferences, and revenue generation. By providing intuitive visualizations and KPIs, the project supports data-driven decision-making and helps improve customer satisfaction and operational efficiency.

This project demonstrates the effectiveness of Business Intelligence tools in solving real-world business problems and highlights the importance of data visualization in modern analytics.

References

- Tableau Public Documentation
- Tableau Desktop User Guide
- Microsoft Excel Documentation
- SmartBridge Internship Project Guidelines
- Business Intelligence and Data Visualization Resources